

Curcuminoids enhance amyloid-beta uptake by macrophages of Alzheimer's disease patients.

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Treatment of Alzheimer's disease (AD) is difficult due to ignorance of its pathogenesis. AD patients have defects in phagocytosis of amyloid-beta (1-42) (Abeta) in vitro by the innate immune cells, monocyte/macrophages and in clearance of Abeta plaques [5]. The natural product curcuminoids enhanced brain clearance of Abeta in animal models. We, therefore, treated macrophages of six AD patients and 3 controls by curcuminoids in vitro and measured Abeta uptake using fluorescence and confocal microscopy. At baseline, the intensity of Abeta uptake by AD macrophages was significantly lower in comparison to control macrophages and involved surface binding but no intracellular uptake. After treatment of macrophages with curcuminoids, Abeta uptake by macrophages of three of the six AD patients was significantly ($P < 0.001$ to 0.081) increased. Confocal microscopy of AD macrophages responsive to curcuminoids showed surface binding in untreated macrophages but co-localization with phalloidin in an intracellular compartment after treatment. Immunomodulation of the innate immune system by curcuminoids might be a safe approach to immune clearance of amyloidosis in AD brain.